

You have been asked to see a previously well 64-year-old man in the PACU with SpO₂ of 85% two hours after laparoscopic right partial nephrectomy during which he lost 1 litre of blood. List the likely causes of the desaturation and outline your approach to managing the patient's hypoxaemia. (From ANZCA 2014B)

Examiner Report

43.4% of candidates passed this question.

As a minimum answers needed to

In part A : Likely cause for desaturation - needs to mention sensible reason for:

- Hypoventilation
- V/Q mismatch
- Poor peripheral perfusion

Part B : Management of hypoxia – need to mention

- Evaluation of breathing and appropriate treatment including potential for re-intubation, PEEP, drainage of pneumothorax, and Chest Xray.
- Evaluation of circulation and level of consciousness and if needed supportive treatment
- Communication with surgeon

Sample Answer

(a)

The low SpO₂ reading of 85% may be related to hypoventilation, V/Q mismatch, or poor peripheral perfusion. Likely causes for his desaturations include:

- Anaesthesia-related
 - Atelectasis/ Mucus plugging
 - Laryngospasm triggered by secretions/ pain
 - Bronchospasm from delayed allergic reaction
 - Hypoventilation secondary to opioids, residual anaesthetics
 - Inadequate NMB reversal/ residual neuromuscular block
 - Upper airway obstruction from laryngeal oedema post-intubation
 - Negative pulmonary pressure oedema (if patient bit on ETT during extubation)
 - Pulmonary oedema from fluid overload (CCF/ renal failure post-nephrectomy).
 - TRALI/TACO if blood transfusion (esp FFP) has been administered
 - Aspiration pneumonitis
- Surgery-related
 - Blood loss leading to reduced O₂ carrying capacity, severe anaemia interfering with pulse oximetry reading.
 - Subcutaneous emphysema/ pneumothorax
 - Splinting of diaphragm due to residual pneumoperitoneum
 - CO₂ embolism from CO₂ insufflation during pneumoperitoneum
- Patient-related
 - Exacerbation of underlying undiagnosed condition - *patient was "previously well"*
 - First presentation of asthma/ COPD exacerbation/ CCF/ neuromuscular disorders
 - Upper airway obstruction from OSA
 - Respiratory tract infection
 - Pulmonary embolism (pro-thrombotic state from underlying malignancy, compounded by long surgery)
 - CNS depression leading to hypoventilation
 - Stroke, post-operative cognitive dysfunction, hypoglycemia, hypothermia
 - Reduced cardiac output leading to hypoperfusion
 - Myocardial ischemia, heart failure, arrhythmias
- Other causes
 - Equipment error (artefactual due to cold peripheries/ shivering)

(b)

This is a concerning situation. I would call for help (alert senior, get trained assistants) and attend immediately to this patient.

This patient has **significant hypoxaemia** 2h post-nephrectomy. I would simultaneously resuscitate patient's ABCs while assessing for possible causes of desaturation. I would also review the rest of vitals (BP, HR, ECG) and confirm pletysmograph waveform to exclude erroneous readings.

Airway	Check and maintain airway patency with <u>head-tilt chin lift</u> manoeuvre, KIV <u>nasal airway</u> , KIV <u>suctioning of upper airway secretions</u> as required. If GCS <8, prepare for re-intubation.
Breathing	Give <u>100% FiO₂ via NRM</u> . Support with <u>bag-mask ventilation</u> followed by re-intubation if inadequate respiratory efforts/ poor oxygenation despite FiO ₂ 100%.
Circulation	Feel for pulse. Look at ECG monitor for <u>ischaemic changes</u> and <u>arrhythmias</u> Run IV crystalloids fast via large bore IV cannula if patient is <u>hypotensive</u>

Evaluate **causes** for desaturations.

- Review history and anaesthesia notes
 - Any significant medical history (OSA/asthma/ COPD/ CCF/ neuromuscular disease/ myopathy) or allergies?
 - Check anaesthesia charts for doses and timings for opioids, sedatives, drugs which patient is allergic to, neuromuscular blocking agents and reversal agents.
 - Check PCA pump for opioid doses administered.
- Physical examination
 - Auscultate chest to exclude tension pneumothorax (absent breath sound, tracheal deviation, hyperresonance to percussion) , anaphylaxis/ bronchospasm (wheeze), pulmonary oedema/ atelectasis (creps).
 - Palpate chest wall for subcutaneous emphysema.
 - Angioedema, rash (allergic/ blood transfusion reaction)
 - Check for surgical dressing and drain for evidence of ongoing bleeding.
 - Check TOF ratio (should be >0.9) using a nerve stimulator.
 - Assess respiratory rate, and check for pupillary size/ response to light (respiratory depression, constricted pupils in opioid overdose).
 - Assess GCS, and neurological status (stroke/ oversedation).
- POCT
 - Target POCUS to evaluate heart and lung pathologies.
 - ABG for respiratory status, and haemoglobin level (retroperitoneal blood loss).
 - Portable CXR (for pneumothorax, aspiration pneumonitis, pulmonary oedema)
 - 12-lead ECG if suspected myocardial ischaemia.
 - Capillary blood glucose, to check and correct hypoglycemia
- CT Brain if there is suspicion of stroke (GCS drop, focal neurological deficits).

Treatment should be directed towards underlying causes for desaturation.

- Encourage deep breathing to open up collapsed alveoli if patient is maintaining spontaneous ventilation.
- Maintain adequate oxygenation by providing supplementary O₂. Bronchodilators for bronchospasm.
- Maintain normotension with fluids / vasopressors / inotropes. Inform surgeon for evaluation and transfuse PCT if ongoing bleeding is suspected.
- Treat residual anaesthesia drug effects with IV Naloxone titrated doses (0.04mg) for opioid overdose, IV Flumazenil for Benzodiazepine overdose, additional reversal agents if NMB was inadequately reversed.
- Intubate to secure airway and for control of ventilation if required. Lung recruitment manoeuvres and PEEP for treatment of atelectasis if patient is re-intubated.
- Needle decompression if tension pneumothorax, followed by chest tube insertion.

Documentation and critical incident reporting should be done up when patient is stable. Surgeon and relatives should be informed. Patient should be sent to High Dependency Unit for monitoring, or ICU for ventilatory support.